

PRINCETON, MN, USA

WMO: 726682

Lat: 45.560N

Lon: 93.608W

Elev: 979

StdP: 14.18

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 54914

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.7	-9.5	-23.7	1.5	-14.1	-17.7	2.2	-8.8	21.0	16.7	19.3	17.5	3.6	280	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	90.0	72.6	86.4	70.9	83.6	69.5	75.8	84.8	73.8	82.9	71.7	80.6	10.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.6	125.5	81.8	71.5	120.7	80.6	69.6	112.7	78.2	39.9	85.0	38.0	82.3	36.2	80.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	18.1	16.2	DB	-24.5	94.1	5.5	2.8	-28.4	96.0	-31.7	97.7	-34.8	99.2	-38.8	101.2	(4)
(5)			WB	-24.8	78.8	5.4	2.0	-28.7	80.2	-31.9	81.4	-34.9	82.5	-38.9	84.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	13.7	17.3	30.3	44.9	56.5	66.1	71.1	68.1	60.7	46.8	32.8	19.2		
	DBStd	22.37	13.45	12.71	12.69	9.88	8.54	6.95	5.91	5.75	8.65	9.21	10.60	12.50		
	HDD50	4563	1126	915	618	210	32	1	0	0	13	172	522	954		
	HDD65	8146	1591	1335	1075	606	290	69	14	32	183	567	965	1419		
	CDD50	2408	0	0	8	57	232	483	653	560	334	74	7	0		
	CDD65	518	0	0	1	3	25	101	203	127	54	4	0	0		
	CDH74	5292	0	0	3	42	369	1004	2076	1235	519	44	0	0		
	CDH80	1609	0	0	0	9	115	277	712	352	140	4	0	0		
	WSAvg	6.5	6.4	6.6	7.0	7.9	7.4	6.2	5.4	5.1	6.1	6.7	6.7	6.4		
	PrecAvg	29.6	0.8	0.8	1.6	2.7	3.2	4.5	4.0	3.8	3.1	2.8	1.5	0.9		
(15) Precipitation	PrecMax	37.8	2.0	1.8	3.3	7.8	6.3	8.3	7.0	8.0	6.4	5.6	4.6	2.2		
	PrecMin	22.2	0.1	0.1	0.7	0.6	1.0	2.2	1.3	0.6	0.7	1.3	0.0	0.2		
	PrecStd	4.8	0.5	0.4	0.8	1.5	1.5	1.6	1.5	1.7	1.7	1.3	1.3	0.5		
	PrecStd	4.8	0.5	0.4	0.8	1.5	1.5	1.6	1.5	1.7	1.7	1.3	1.3	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	52.3	69.8	79.4	90.2	90.8	93.2	91.2	89.6	79.4	65.9	47.6		
		MCWB	38.2	46.0	56.4	59.4	67.3	73.5	74.0	73.0	71.2	63.2	53.8	44.7		
	2%	DB	38.7	44.6	61.0	72.4	82.1	86.2	90.2	87.9	83.9	72.8	59.3	42.7		
		MCWB	35.0	39.0	51.6	53.3	63.3	69.9	73.8	72.4	69.4	60.0	50.6	37.8		
	5%	DB	35.8	38.8	54.4	67.6	78.6	82.3	87.7	83.6	80.8	66.4	53.9	37.2		
		MCWB	33.5	35.4	47.4	52.2	61.0	67.0	72.4	70.3	67.3	56.3	46.9	34.8		
	10%	DB	32.0	35.9	47.8	63.1	72.7	80.8	83.8	81.2	75.3	63.0	47.9	35.8		
		MCWB	30.6	33.2	42.4	49.5	58.2	66.0	70.8	67.9	64.7	53.8	42.7	33.9		
	0.4%	WB	38.6	46.0	60.0	61.7	71.8	76.0	78.9	77.2	75.4	67.1	57.6	44.1		
		MCDB	43.8	51.5	66.2	76.7	84.2	86.0	88.5	87.5	83.1	74.0	62.7	47.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.5	39.4	52.9	57.3	66.9	72.9	76.5	74.7	71.8	62.1	52.3	37.9		
		MCDB	38.3	43.5	58.8	68.1	76.7	82.5	85.7	83.5	80.1	69.6	58.9	40.7		
	5%	WB	32.9	35.8	47.4	53.5	63.7	70.5	74.6	72.3	69.2	58.4	46.7	35.4		
		MCDB	35.0	39.2	54.7	63.6	72.7	79.9	83.5	80.7	76.7	65.8	52.2	37.5		
	10%	WB	30.2	32.8	42.1	50.4	60.5	68.0	72.5	70.0	66.7	54.5	43.4	33.2		
		MCDB	31.6	35.4	47.8	60.8	69.9	76.8	81.5	77.7	74.3	60.8	48.3	35.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	18.8	19.1	21.5	22.6	21.1	21.8	22.2	22.4	20.0	16.9	15.2		
		MCDBR	18.2	18.9	25.5	29.9	30.5	24.7	24.5	25.1	25.7	27.6	24.8	16.7		
		MCWBR	14.9	14.2	16.5	15.9	14.3	11.5	11.0	11.8	13.2	15.6	16.2	13.6		
		MCDBR	17.2	17.7	24.2	25.1	24.9	21.0	21.0	21.2	21.5	22.5	22.4	15.1		
	5% WB	MCWBR	14.6	13.4	16.3	15.0	13.6	11.5	11.2	11.7	12.8	14.9	15.6	12.6		
		MCWBR	14.6	13.4	16.3	15.0	13.6	11.5	11.2	11.7	12.8	14.9	15.6	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.273	0.313	0.372	0.406	0.392	0.386	0.398	0.370	0.316	0.286	0.262		
	taud		2.397	2.366	2.375	2.292	2.249	2.333	2.373	2.341	2.392	2.538	2.508	2.425		
	Ebn at Noon		272	291	292	281	274	278	278	270	269	272	262	260		
	Edh at Noon		23	28	33	39	42	39	37	37	32	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		539	885	1231	1474	1694	1875	1986	1655	1278	799	505	374		
	RadStd		61	84	107	136	137	137	67	124	88	108	45	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page